UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 008 · Problems on Taylor's and Maclaurin's series · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

Contents

1. Figure G1.1 · $\sin(xy)$ near $(1, \pi/2)$: a slice at $y = \pi/2$
2. Figure G2.1 · $\log(2x + y + 1)$ as a function of $t = 2x + y$
3. Figure G3.1 · Approximating $\tan^{-1}(xy)$ at $(0.9, -1.2)$

Colours follow the Okabe–Ito order: blue = the function, vermilion = first-degree polynomial, green = second degree, purple = third degree; grey dashed lines are references. The deck has no graphs, so all figures are added for clarity and drawn from the deck's own functions.

1 Taylor expansion about a point (a, b)

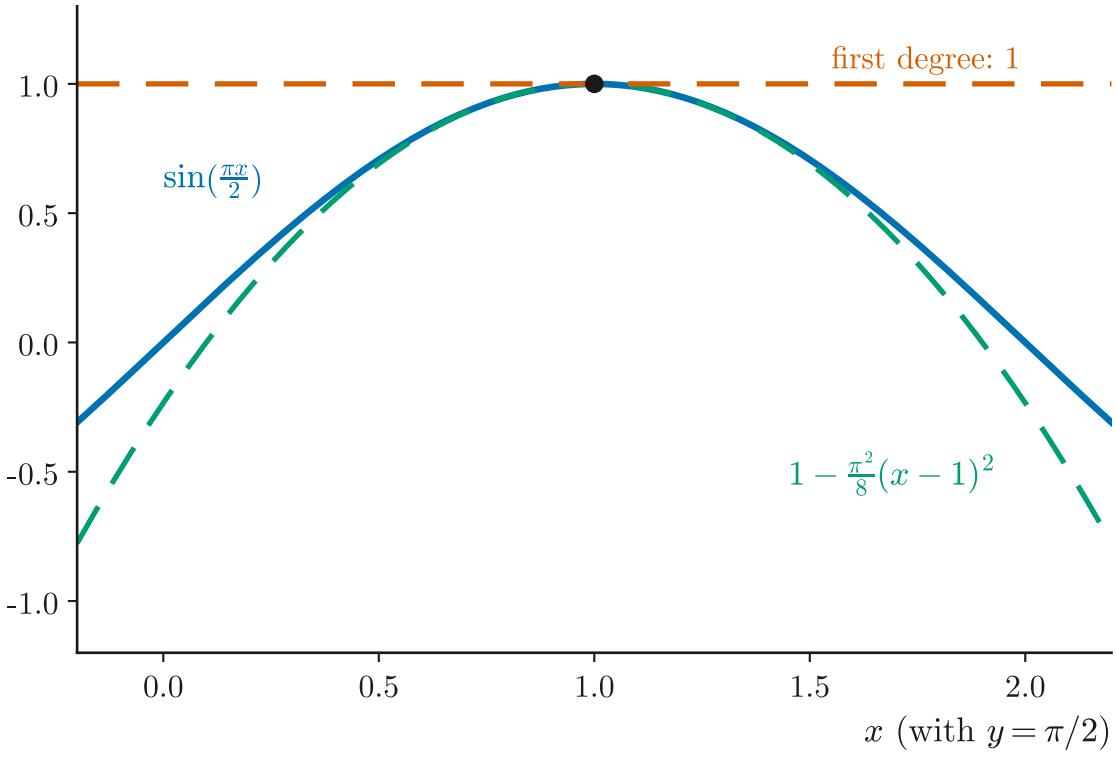


Figure G1.1. $\sin(xy)$ near $(1, \pi/2)$: a slice at $y = \pi/2$. ESA ★

ADDED FOR CLARITY S008·14 · S008·16

How to read it. Along $y = \frac{\pi}{2}$, $\sin(xy) = \sin\frac{\pi x}{2}$ (blue) has a maximum at $x = 1$, so both first partials vanish and the first-degree polynomial is the flat line 1 (vermilion). The second-degree polynomial $1 - \frac{\pi^2}{8}(x-1)^2$ (green) follows the curvature near $x = 1$.

2 Maclaurin expansion about the origin

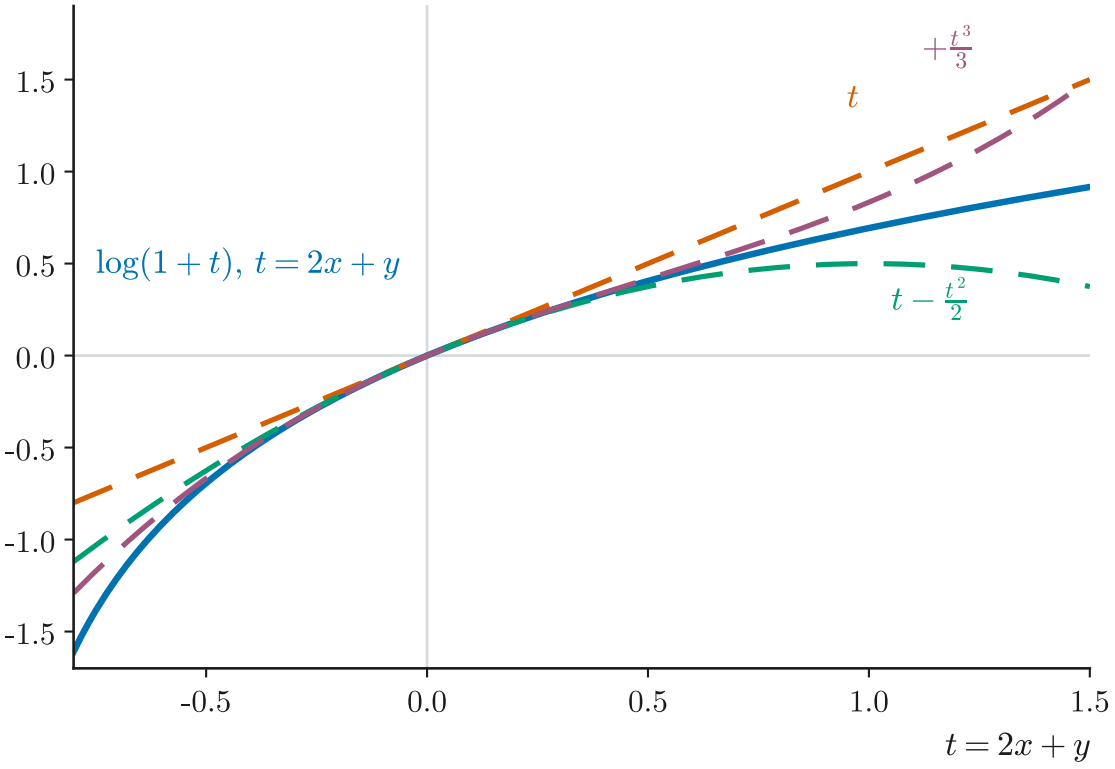


Figure G2.1. $\log(2x + y + 1)$ as a function of $t = 2x + y$. ESA ★★

ADDED FOR CLARITY S008·5 · S008·6

How to read it. The function depends only on $t = 2x + y$, so its Maclaurin polynomials are t, $t - \frac{t^2}{2}$ and $t - \frac{t^2}{2} + \frac{t^3}{3}$. Each extra degree follows $\log(1 + t)$ (blue) further from $t = 0$; all fail as $t \to -1$, where the logarithm blows up.

3 Approximate values from a Taylor polynomial

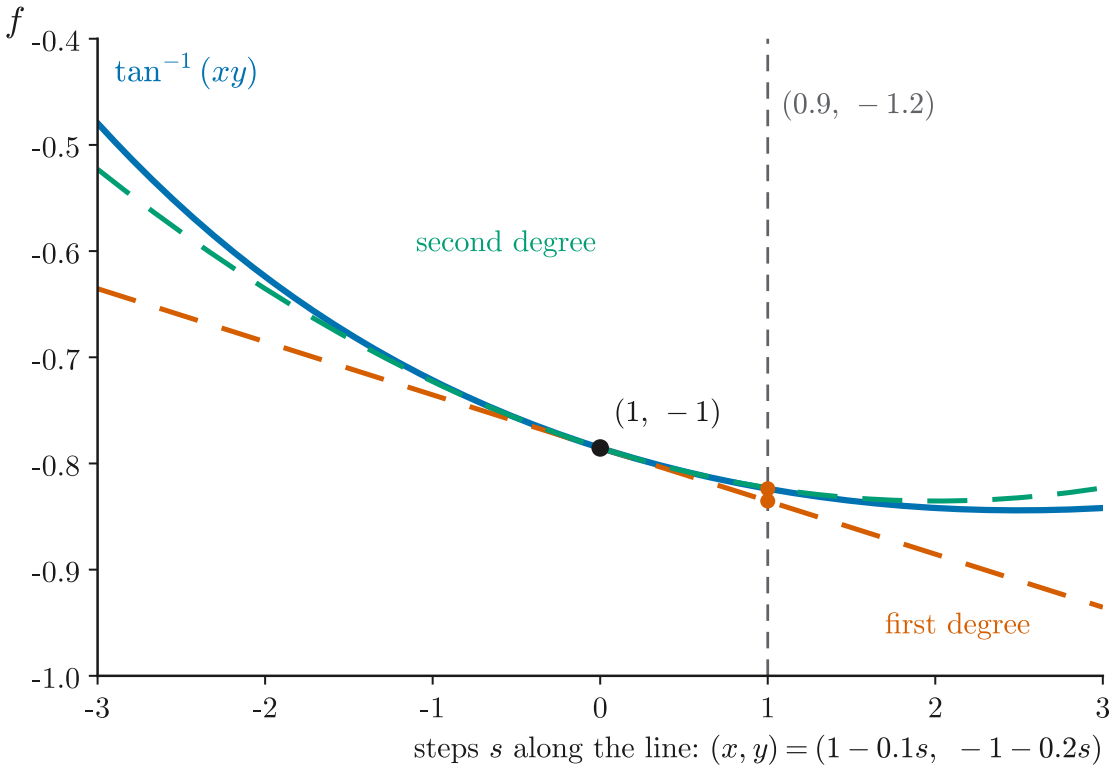


Figure G3.1. Approximating $\tan^{-1}(xy)$ at $(0.9, -1.2)$. ESA ★★

ADDED FOR CLARITY S008·7 · S008·9 · T008 00:14:18

How to read it. Values along the straight line from $(1, -1)$ (black dot, $s = 0$) through $(0.9, -1.2)$ ($s = 1$, grey dashed line). At $s = 1$ the first-degree polynomial (vermilion) gives -0.835, while the function (blue) is -0.824; the second degree (green) is almost on the curve there.